Author & Year	Study Design	Population	Intervention vs Comparator	Key Outcomes Measured	Key Findings	Limitations
Laffel LM et al., 2023	Randomized controlled trial (Phase 3)	Youth with Type 2 diabetes, ages 10-17	Empagliflozin vs Linagliptin vs Placebo	HbA1c change at 26 weeks	Empagliflozin reduced HbA1c by 0.84% vs placebo (p=0.012); linagliptin did not achieve significance (p=0.29). SGLT2 superior on glycemic control in youth.	Youth population only; placebo-controlled not head-to-head; funded by Boehringer Ingelheim and Eli Lilly
Khan F et al., 2024	Systematic review of RCTs	Type 2 diabetes patients	SGLT2 vs DPP-4 vs metformin vs insulin	HbA1c, cardiovascular outcomes, renal benefits, weight, hypoglycemia risk	SGLT2 inhibitors superior on cardiovascular and renal outcomes including reductions in major adverse cardiovascular events and kidney disease progression. DPP-4 inhibitors well tolerated but efficacy diminishes over time.	Includes multiple drug comparators beyond SGLT2 vs DPP-4; not a direct head-to-head comparison
Mesri Alamdari N et al., 2025	Randomized double-blind clinical trial	Type 2 diabetes patients on combination therapy	Empagliflozin vs Sitagliptin vs Metformin	HbA1c, fasting blood sugar, weight, blood pressure, lipid profile	Empagliflozin produced greater reductions in HbA1c (-1.8% vs -1.35%), weight (-4.1kg vs metformin), blood pressure, and triglycerides compared to sitagliptin and metformin. SGLT2 superior on cardiometabolic outcomes.	No direct statistical comparison between empagliflozin and sitagliptin; all three arms compared against metformin separately
Au PCM et al., 2022	Real world evidence — retrospective cohort study	Type 2 diabetes patients, Hong Kong electronic health records, n=33,320	SGLT2 inhibitors vs DPP-4 inhibitors	Pneumonia incidence and pneumonia mortality	SGLT2 associated with 29% lower pneumonia risk and 43% lower pneumonia mortality vs DPP-4 inhibitors over 3.8 year follow-up	Hong Kong population — may not generalize to US patients; pneumonia not a primary diabetes outcome
Yao X et al., 2026	Real world evidence — retrospective cohort study	100 Type 2 diabetes patients (50 per group), China	SGLT2 inhibitors vs DPP-4 inhibitors	HbA1c, major adverse cardiovascular events, weight, blood pressure, renal function	Comparable HbA1c reduction between groups. SGLT2 patients had significantly fewer major adverse cardiovascular events (2% vs 16%, $p<0.05$) and greater improvements in weight, blood pressure, and kidney function.	Small sample size of 100 patients; Chinese population may limit US generalizability
Choe HJ et al., 2026	Real world evidence — retrospective switch study with propensity score matching	168 Type 2 diabetes patients switching between drug classes (84 per group), South Korea	Switching to SGLT2 inhibitors vs switching to DPP-4 inhibitors	HbA1c, fasting plasma glucose, BMI, liver enzymes, renal function	Switching to DPP-4 produced greater HbA1c reduction. Switching to SGLT2 improved fasting glucose and liver metabolic parameters. Neither drug universally superior — benefits depend on patient characteristics.	Small sample of 168 patients; Korean population; switch study design differs from treatment initiation studies
Pawaskar M et al., 2021	Cost-effectiveness analysis — US payer perspective	Type 2 diabetes patients not at HbA1c target on metformin plus DPP-4 therapy	Adding SGLT2 inhibitor to DPP-4 vs switching to GLP-1 receptor agonist	Life years, QALYs, total costs, cardiovascular and renal complications	Adding SGLT2 to DPP-4 therapy dominated switching to GLP-1 — marginally better outcomes with $9,511 lifetime cost savings per patient	Comparison vs GLP-1, not standalone DPP-4; model-based analysis using published inputs rather than primary data
Kongmalai T et al., 2024	Cost-utility and budget impact analysis	Type 2 diabetes patients with heart failure, Thailand national database	SGLT2 inhibitors vs standard treatment	QALYs, ICER, heart failure hospitalizations, budget impact over 5 years	SGLT2 inhibitors reduced heart failure hospitalizations and improved QALYs. ICERs ranged from $4,632 to $48,952 per QALY depending on specific drug. Not cost-effective at Thailand willingness-to-pay threshold of $4,564 per QALY.	Thailand healthcare setting — costs and willingness-to-pay thresholds not applicable to US payers; heart failure subpopulation only
Morton JI et al., 2023	Cost-effectiveness analysis — microsimulation model	1.1 million Type 2 diabetes patients, Australian Diabetes Registry	Widespread SGLT2 inhibitor use vs current use vs GLP-1 receptor agonists	QALYs, total costs, ICER, cardiovascular events, kidney disease progression 2020-2040	Widespread SGLT2 use cost-effective at AU$23,717 per QALY gained, well below AU$28,000 willingness-to-pay threshold. GLP-1 RAs not cost-effective at current prices. SGLT2 generated 176,446 QALYs over 20 years.	Australian healthcare perspective — costs and thresholds differ from US setting; GLP-1 comparator not DPP-4
Htoo PT et al., 2024	Real world evidence — propensity score matched cohort study, US Medicare data	23,335 matched pairs, US Medicare beneficiaries age 65+, mean age 72 years	Empagliflozin vs DPP-4 inhibitors	Hospitalizations, ED visits, inpatient days, office visits, total healthcare costs PMPY	Empagliflozin associated with 14% fewer hospitalizations, 14% fewer ED visits, and $1,109 lower total cost PMPY vs DPP-4. Drug costs higher for empagliflozin but offset by lower inpatient and outpatient costs. Larger cost advantage in CVD patients (-$2,005 PMPY).	Medicare population age 65+ may not represent younger diabetes patients; study period 2014-2018 predates recent SGLT2 price changes